

Computer-Aided Design and Analysis

Preamble

This report aims to produce a 3D model of a car rim and analyse stress factors on the component using Finite Element Analysis (FEA). The chosen software is Fusion 360 due to its ease of workflow and flexibility in designing relatively small projects (Gibbons, 2020). The objective of this report is to also find a solution to extreme conditions and loads that the wheel needs to withstand from a design perspective. Each concept is investigated and validated using the chosen CAD software followed by an explanation of results. Key takeaways and improvements are then discussed, leading to a conclusion.

Design

This section outlines the design process undertaken to successfully generate the wheel rim using the correct material. CAD drawings with dimensions have been provided to validate all the steps.

The general structure of modelling the rim on Fusion 360 is listed below:

1. Start an initial sketch on the top plane as shown in figure 1

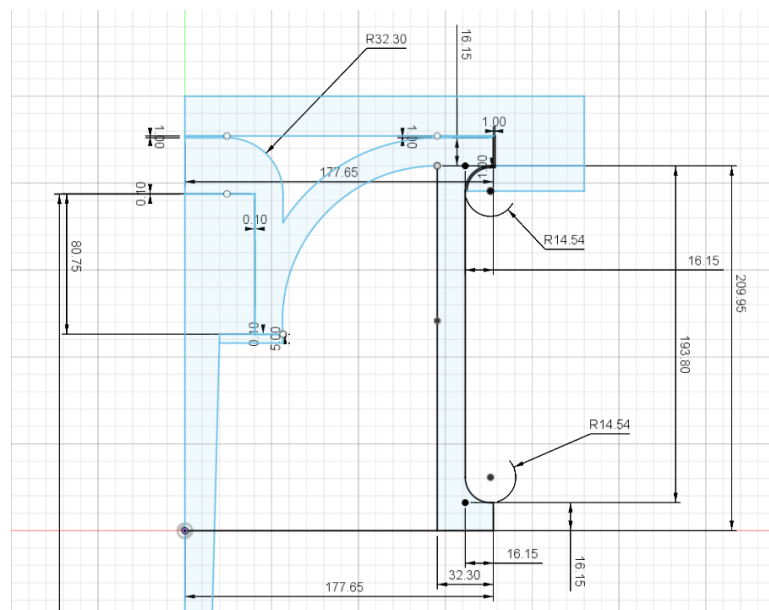


Fig 1: Initial top plane sketch

2. Ensure sketch dimensions accurately meet the requirements. This sketch also contains the movement armature and fixed fixture which will be used later for FEA
3. Revolve sketch around centre axis (Y in this case) to produce the 3D model shown in Appendix 1

4. Start another sketch on the centre bore plane
5. Use circular pattern to mirror spokes with respect to the centre as shown in Appendix 2
6. Ensure sketch dimensions meet requirements
7. Use extruded cut to obtain final car rim model as shown in figure 3
8. Additional profile – Model the armature and fixture around the wheel using the offset tool as shown in figure 4



Fig 2: Wheel rim render

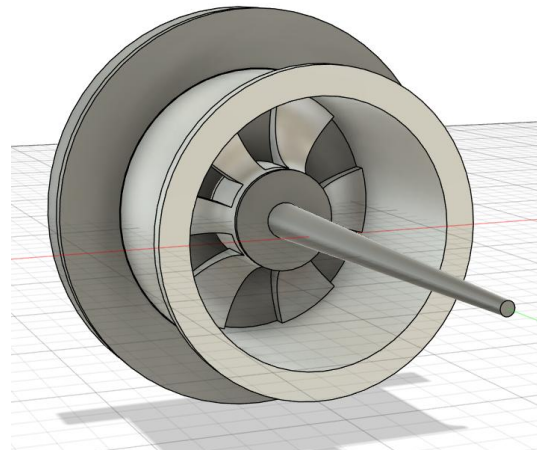


Fig 3: Fixed boundary assembly

9. Assign material to each body: Aluminium alloy 2014 for the car rim and steel for the armature and fixture. Steel is the default material on Fusion 360 and is a close approximation of a real-world car axle

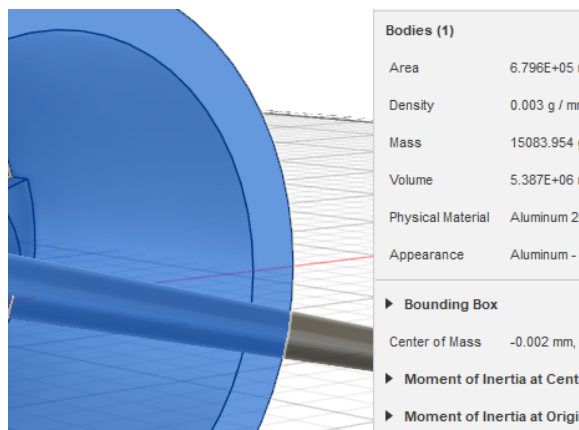


Figure 4: Wheel material assignment

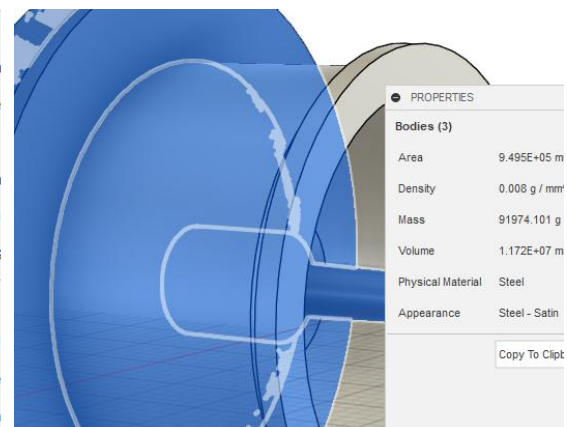


Figure 5: Boundary material assignment

Once the required model has been generated, a first/third angle orthogonal projection drawing is created as shown in figure 5. This is used to verify that all dimensional requirements are met.

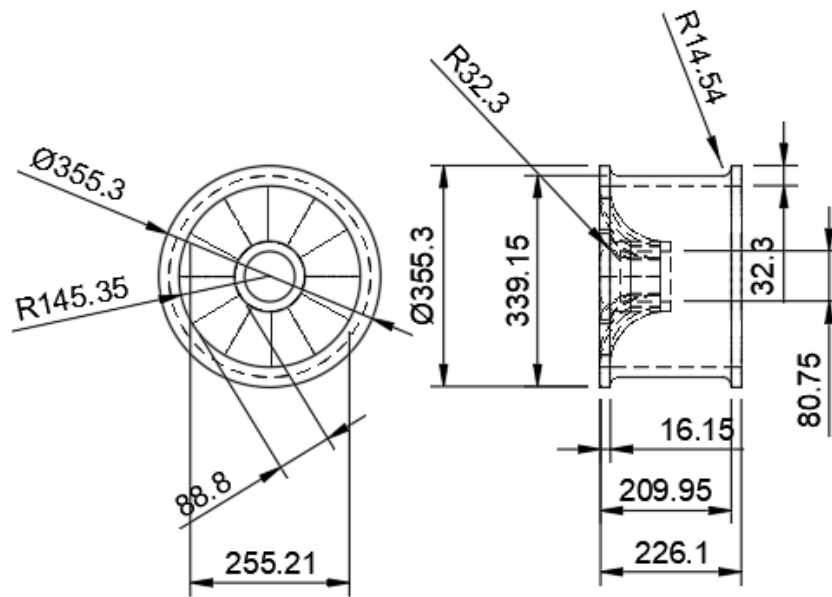


Figure 6: Orthogonal Projection

Analysis (FEA)

This section is aimed at carrying out fatigue and stress analysis on the wheel rim modelled in the previous section. Throughout all analyses, the magnitude of the force and the distance at which the force acts upon are kept constant (control variable).

Safety factor is a ratio of the specific strength of a material to the applied force on the material. It is another way of describing the maximum stress a material or component can withstand before it becomes ineffective in function. (Post, 2024)

Mechanical stress is defined as the force acting on a unit area.

$$\sigma = \frac{F}{A}$$

The von Mises stress is an average of all local normal and shear stresses acting on a component. This is the default stress distribution scale on Fusion and therefore required no modifications in terms of simulation settings. (Simscale, 2023)

The safety factor and von Mises stress plots are the main points of comparison in this report. The cornering fatigue test requires a remote load of 5000N to be applied 500mm from the wheel hub. Two different analysis approaches were carried out.

For the first study, a fixed structural constraint was applied to the fixture while a force of 5000N was applied to the end of the armature in the x-direction. Automatic contacts were generated with a tolerance of 0.5mm.

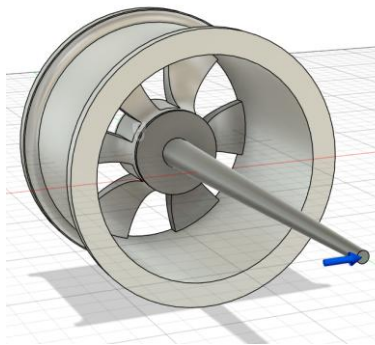


Figure 7: Force applied on armature

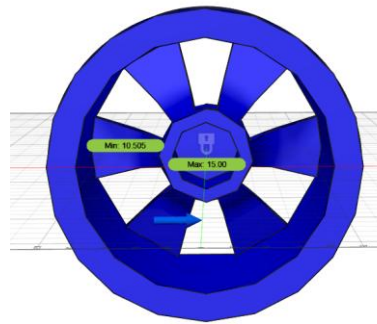


Figure 8: Safety Factor when force is on armature

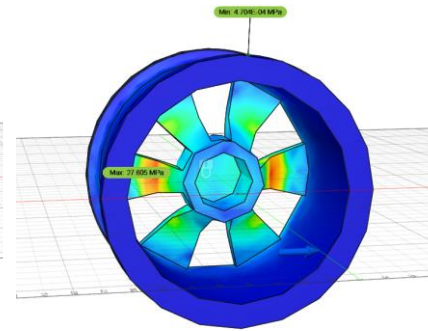


Figure 9: von Mises stress plot when force is on armature

The results obtained are as follows:

Max stress (MPa)	Min Stress (MPa)	Min Safety Factor	Max Safety Factor
27.605	0.00047	10.505	15.00

Table 1

Similar to the first study, for the second study, a fixed structural constraint was applied to the fixture. A remote load of 5000 N was applied to the inner faces of the hub in the x-direction at a distance of 500mm from the centre of the wheel hub in the y-direction. Automatic contacts were generated with a 0.5mm tolerance.

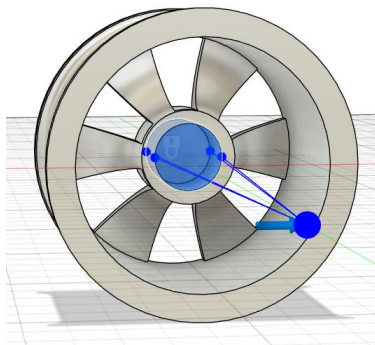


Figure 10: Remote force applied

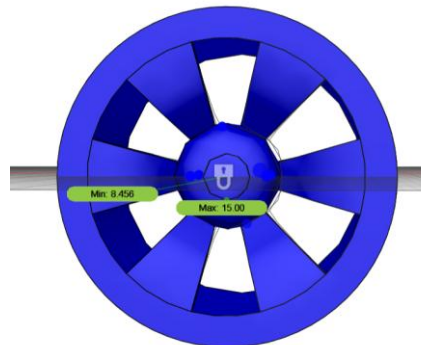


Figure 11: Safety Factor when remote force is used

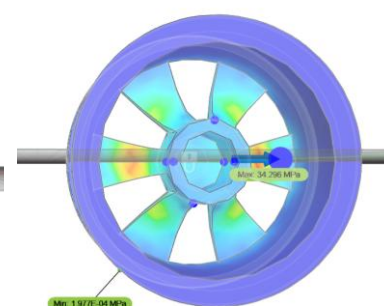


Figure 12: von Mises plot when remote force is used

The results obtained are as follows:

Max stress (MPa)	Min Stress (MPa)	Min Safety Factor	Max Safety Factor
34.296	0.00002	8.456	15.00

Table 2

The reasoning behind this approach is because, on investigation of the first method, it was observed that the armature absorbs a proportion of the force targeted on the wheel. Comparison of safety factor and Von Mises Stress values indicate a strong possibility that this might be the case due to a reduced safety factor for the second study, caused by a greater load across the same area. This is also validated by the increase in maximum stress values.

	MAX STRESS (MPa)	MIN SAFETY FACTOR
STUDY 1	27.605	10.505
STUDY 2	34.296	8.456

Table 3

Based on results, the second approach, using remote force, will be the primary method for all further analyses as it is assumed to be more accurate.

Investigation

As the force in this specific simulation is acting through the x-axis, the two spokes that coincide with the x-axis experience the most stress. The objective is to reduce the safety factor, ideally to a 2:1 ratio. Before any modifications are made, it is important to consider mesh density of the model using initial dimensions.

Mesh density of a surface is defined as the number of nodes or elements per unit area. The greater the mesh density (finer mesh), the more accurate the results. This is because a greater number of points are analysed and considered compared to a low mesh density (coarse mesh). (Liu, 2013)

The initial model had a coarse mesh density as shown in Figure 13. For a more precise analysis, the mesh density must be made finer by decreasing the element size. This can be done by changing the average element size, turn angles and size ratios. The main objective is to find out what value, the safety factor, converges to. Regions with significant stress concentration in the coarse mesh can be studied better using a finer mesh as shown in the stress distribution plots (fig x and x), to then be modified later on if required.

1. Coarse mesh

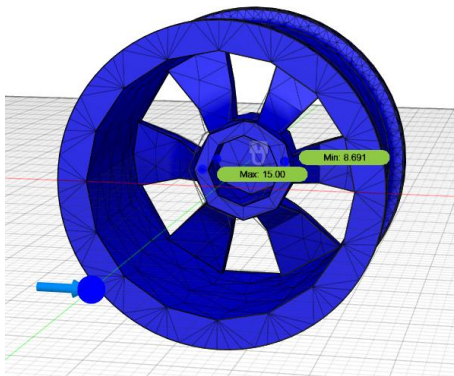


Figure 13: Safety Factor of coarse mesh

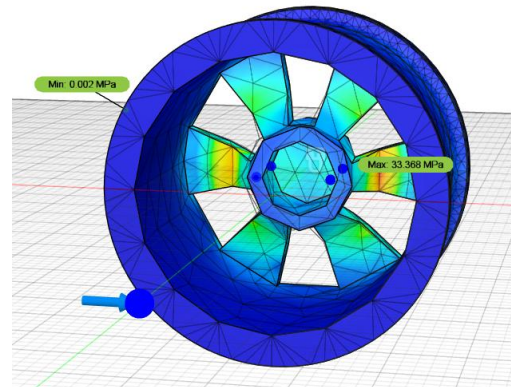


Figure 14: von Mises stress plot of coarse mesh

The results obtained are as follows:

Max stress (MPa)	Min Stress (Mpa)	Min Safety Factor	Max Safety Factor
33.368	0.002	8.691	15.00

Table 4

2. Medium mesh

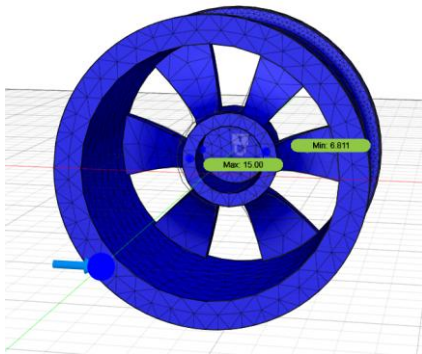


Figure 15: Safety Factor of medium mesh

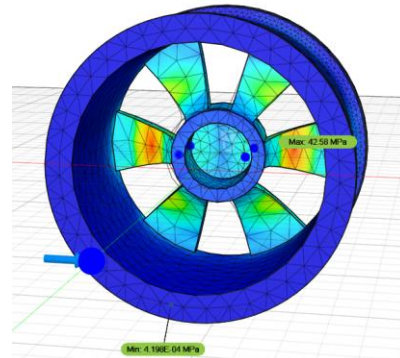


Figure 16: von Mises stress plot of medium mesh

The results obtained are as follows:

Max stress (MPa)	Min Stress (MPa)	Min Safety Factor	Max Safety Factor
42.58	0.0004	6.811	15.00

Table 5

3. Fine mesh

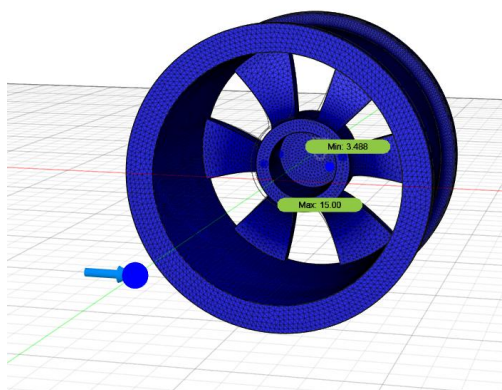


Figure 17: Safety Factor of fine mesh

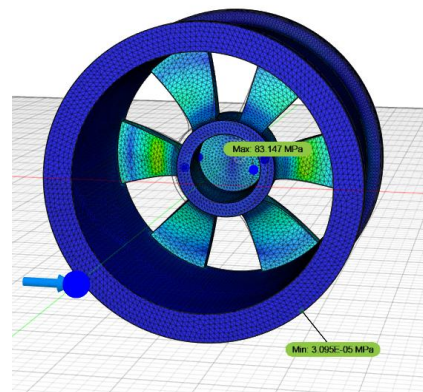


Figure 18: von Mises stress plot of fine mesh

The results obtained are as follows:

Max stress (MPa)	Min Stress (MPa)	Min Safety Factor	Max Safety Factor
83.147	0.00002	3.488	15.00

Table 6

From initial evaluation, it was found that the safety factor does not converge. This may be due to several parameters like material, mesh geometry and limitations of software. Simulations below a scale size of 5% can have artificial results as suggested by Fusion 360. The specific material used (Aluminium 2014 T4), might also not be optimised for such a fine mesh density and hence results may not be accurate. On closer inspection, the safety factors were decreasing non-linearly. Since, absolute deviation, as shown in the table below, does not decrease with variation in mesh density, it is difficult to predict any approximations for a mesh convergence point.

Mesh density	Min Safety Factor (SF)	Mean Safety Factor (SF_{mean})	Absolute Deviation SF_{abs} ($SF_{abs} = SF - SF_{mean} $)
Coarse	8.691	6.33	2.361
Medium	6.811		0.481

Fine	3.488		2.842
------	-------	--	-------

Table 7

Discussion and Improvements

To further reduce the safety factor to the desired 2:1 ratio, adjustments to various parameters such as sketch dimensions, profiles and materials were made.

The current material used is Aluminium Alloy 2014. Aluminium alloy wheels are the most common modern car rims due to its light-weight property when compared to steel. The manufacturing process involves casting and forging which enables complex designs to be constructed. Materials with lower tensile properties are to be considered if the safety factor is required to be reduced. Magnesium alloy, although not ideal for commercial cars due to its flammable nature, is much more brittle compared to aluminium alloys and therefore could potentially match the safety factor requirements. It is used in a lot of performance vehicles and would be a key improvement if cost wasn't a factor. (O'Hop, 2021) The material used was Magnesium AZ91E T6 and a safety factor of 3.692 was obtained on normal mesh density and initial dimensions.

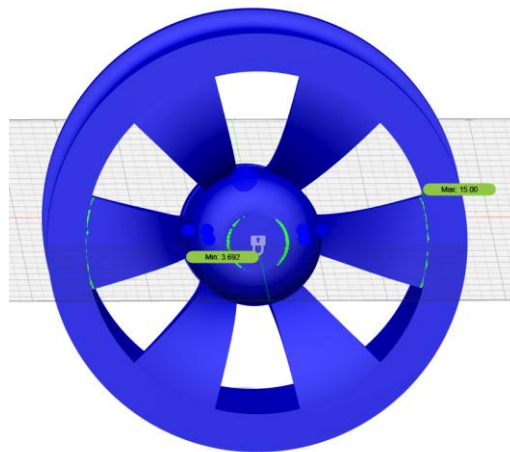


Figure 19: Safety Factor of adjusted material

From previous results, it was concluded that the spokes experience the most stress. Adjusting the sketch profile by adding spokes should therefore reduce safety factor. Hence, a greater number of thinner spokes were added (8 and 10 spokes instead of the stock 6). As a result, a safety factor of 2.4 was achieved. A greater number of spokes with relatively less surface area would be harder to manufacture due to casting complexities and handling requirements.

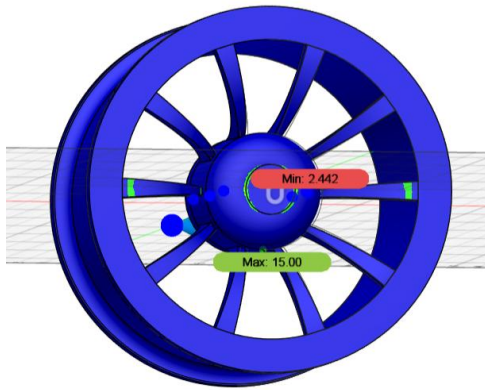


Figure 20: Safety Factor of 10 spokes

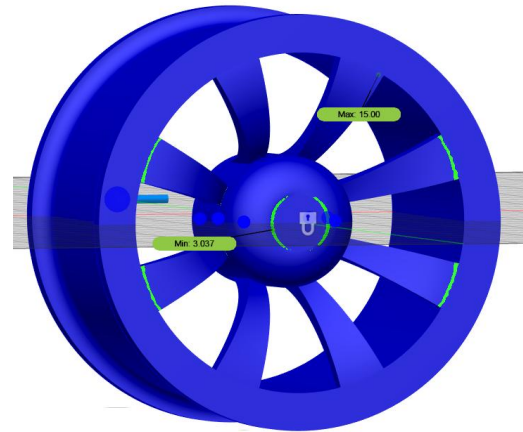


Figure 21: Safety factor of 8 spokes

The results obtained from the analysis showed that the 8-spoke model had a safety factor of 3.037 while the 10-spoke model had 2.442. This is a huge improvement from the initial 6-spoke design.

Adjustments were also made to the wheel lip and spoke width as shown in the figure. Lip size was reduced by a factor of 33% while the diameter of the inner spoke edge was increased by 66%. These modifications helped to bring down the safety factor to 2.8 from 8, which is a significant difference. The main manufacturing methods involved would be the usual casting and machining followed by heat treating and precise finishing to accommodate thinner profiles. (Soori, 2022)

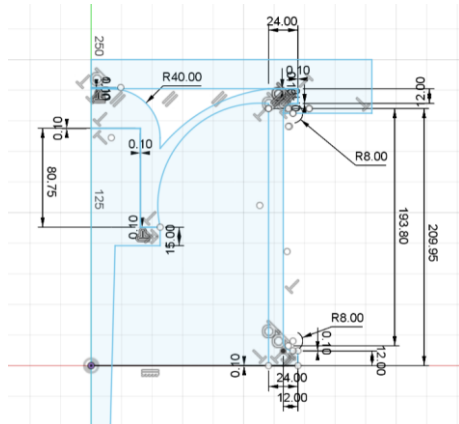


Figure 22: Sketch adjustments

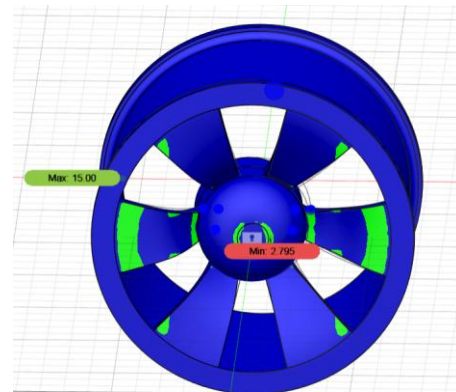


Figure 23: Safety factor of adjusted sketch

Finally, to achieve a safety factor of 2, a combination of these design variations was implemented. Adjustments were made to both the number of spokes and material selection. It was decided to use the Magnesium alloy (Magnesium AZ91E T6) and increase the number of spokes to 10. This brought the safety factor to a target value of 2.02. Additionally, fillets were added to the ends of the spokes to help reduce stress concentration.

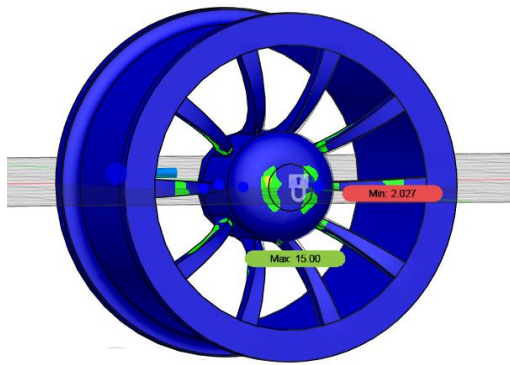


Figure 24: Safety factor of combination

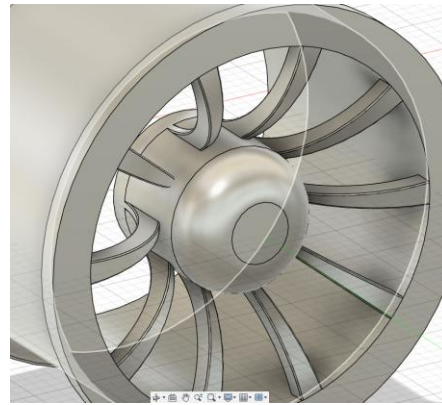


Figure 25: Filleted spokes

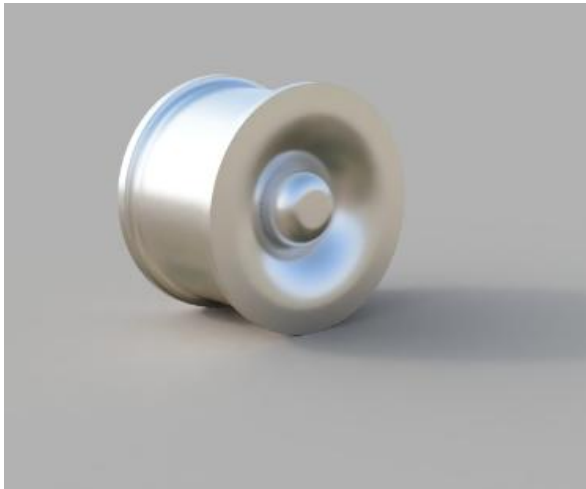
Conclusion

In summary, this report demonstrates the design and analysis of a standard wheel rim along with recommendations for potential improvements. Key observations and manufacturing methods have been discussed where relevant. ISO 14400:2021 standards have been considered during all the steps of the design process. Structural integrity and ability to endure critical conditions were core aspects of comparison.

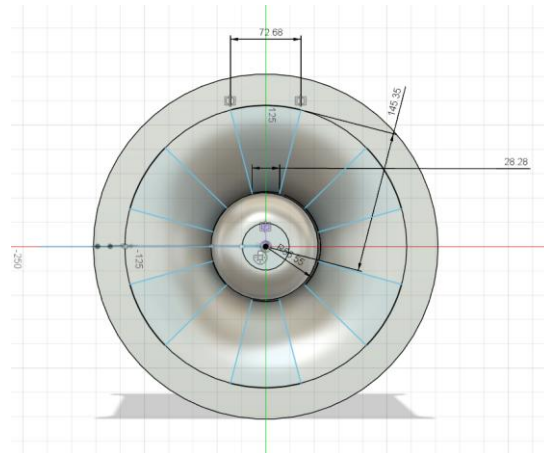
References

- Gibbons, W. (2020, March 21). *Fusion 360 vs Solidworks*. Retrieved from willgibbons: <https://www.willgibbons.com/fusion-360-vs-solidworks/>
- Liu, Y. (2013). *Effects of mesh density on Finite Element Analysis*. South Dakota State University: University of Louisiana.
- O'Hop, H. (2021, July 15). *What are the pros and cons of different wheel materials*. Retrieved from The Drive: <https://www.thedrive.com/cars-101/41434/pros-and-cons-of-different-wheel-materials>
- Post, H. (2024, October 10). *What is Factor of Safety?* Retrieved from Tradesafe: [https://trdsf.com/blogs/news/what-is-factor-of-safety#:~:text=long-term%20durability-,Factor%20of%20Safety%20Formula,\)%20%3D%20Material%20Strength%2FApplied%20Load](https://trdsf.com/blogs/news/what-is-factor-of-safety#:~:text=long-term%20durability-,Factor%20of%20Safety%20Formula,)%20%3D%20Material%20Strength%2FApplied%20Load)
- Simscale. (2023, August 11). *What is von Mises Stress?* Retrieved from Simscale: <https://www.simscale.com/docs/simwiki/fea-finite-element-analysis/what-is-von-mises-stress/>
- Soori, M. (2022). *Manufacturing Process of car Alloy wheels*. Kyrenia, Turkey: HAL Open Science. Retrieved from Turbo Tech.

Appendices



Appendix 1: Revolved sketch



Appendix 2: Centre Bore sketch